



ರೈ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
Rai Technology University
(Estd. under Karnataka Act. No. 40 of 2013)

ASSIGNMENT -1

Subject Title: Human Computer Interaction

Subject Code: OCS343T

Submitted By:

Student Name: Pavan Kumar

University Enrolment Number: RTU24101CS039

Program Name: B.TECH CSE (AIML)

Semester: 4TH Sem

Session: 2026-2027

Submitted To:

Faculty Name: Mr Abhishek Gupta

Department of Computer Science & Engineering College

Of Engineering

Rai Technology University

HCI in The Software Process

Introduction

Human computer interaction (HCI) is the study of how people interact with the computers. It focuses on making systems easy to use and understand.

It is not only about technology but also about human behaviour. A good system is not just functional, it is also usable. HCI helps in designing systems that people can use without confusion.

It connects human needs with system design.

Importance of HCI

HCI improves the overall user experience.

It makes systems simple and intuitive.

Users can complete tasks faster. It

reduces errors during interaction.

It increases user satisfaction. A well designed system builds trust. poor design

leads to frustration. HCI helps avoid

that problem. It also reduces development

cost in the long run.

HCI in software Development life Cycle.

HCI is used in every stage of development. It starts from understanding user needs. Developers study how users think and behave. Design is created based on user expectations. Interfaces are planned carefully.

During development, simplicity is maintained. Testing includes real user test is conducted. Feedback is collected and improvements are made. HCI continues even after deployment.

Reviews are collected from app markets based on which new versions of the software are updated.

Usability Engineering

Usability engineering focuses on ease of use. It ensures that users can complete tasks properly. It measures efficiency and satisfaction. Systems should be easy to remember. User errors should be minimised. Performance is continuously evaluated. Improvements are made based on results.

Iterative Design

HUI follows an iterative process. Design is not done only once. It is repeated and improved. First a Design is created. Then it is tested. problems are identified. changes are made. This cycle continues until the system improves.

Prototyping

Basic Concept

- Prototyping is the process of creating an early version of a software system.
- It is used to test ideas before the final system is developed.
- A prototype helps visualise how the system would look and work.

Flexibility

- Prototypes are easier to modify compared to completed systems.
- Changes can be made quickly without high cost or effort.
- This reduces the chances of major errors in later stages -

Design Principles

Consistency is importance in design.

Similar actions should give similar results.

Feedback should always be prepared.

User should know what's happening.

Simplicity should be maintained. Complex

design confuses users. Important elements

should be visible. Errors should be

prevented. Systems should support different types of users.

Design Rationale

Design rationale explains why decisions are made. It helps in understanding design choices. It improves communication in teams.

Conclusion

HUI plays an important role in software development. It focus on user needs and behavior. It improves usability and efficiency.

It reduces errors and increases satisfaction. It is used in every stage of development. A system without HUI is difficult to use. A system with HUI is effective and user-friendly.